



University of Al-Ameed / College of Medicine
Department of Physiology and Medical Physics



Introduction to blood Physiology

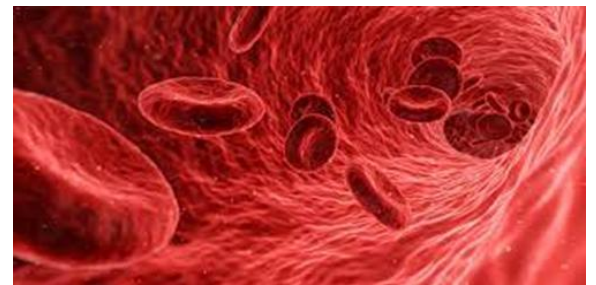
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M. Muntadher Najm
2026-2027

Introduction

- **Circulatory system** is the transport system that supplies O₂ and substances absorbed from gastrointestinal tract to the tissues, returns CO₂ to the lungs and other products of metabolisms to the kidneys, functions in the regulation of body temperature, and distribute hormones and other agents that regulate cell function.

- **Hematology**

Is the study of the blood forming tissues and circulating blood components.



CIRCULATORY SYSTEM

VESSELS OF THE UPPER LIMB

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LUNG VESSELS

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LIVER VESSELS

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CAPILLARIES

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BRAIN VESSELS

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HEART

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BOWEL VESSELS

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RENAL BLOOD VESSELS

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VESSELS OF THE LOWER LIMB

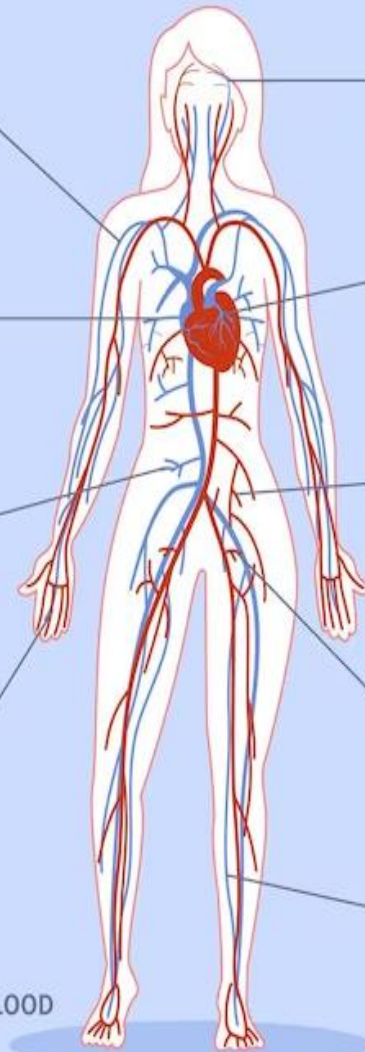
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ARTERIAL BLOOD



DEOXYGENATED BLOOD



Blood

- Is the fluid that circulates in the cardiovascular system, has many kinds of chemicals dissolved in it and millions upon millions of cells floating in it.
- An adult has about 5 liters of blood in his body. The circulating blood accounts for 5-7% of total body weight and is composed of two major elements:



Blood

A) Cellular elements:

1. Red blood cells (Erythrocytes)

- Carrying O_2 & CO_2 between lungs and tissues via hemoglobin **Hb** content in their cytoplasm.
- Red color= occurs when **O_2** is attached to the **Hb**.
- Life span is short because it **lacks nucleus**.

2. White blood cells (Leukocytes)

- Colorless nucleated cell, main function is protection against invading organisms.



Blood

3. Platelets (Thrombocytes)

- Non-nucleated disk shape, prevent leakage or bleeding caused by vessel walls.

B) Fluid Elements (plasma)

- **Plasma** is composed of:

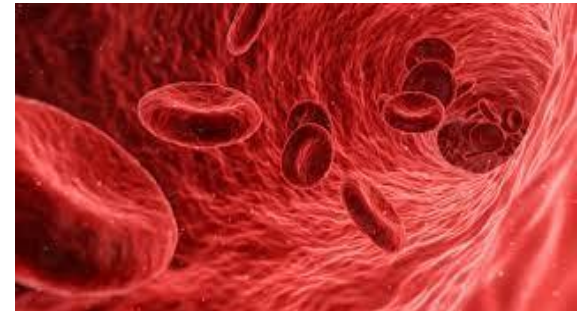
1. **Water 70%** of body
2. **Electrolytes:** essential to cellular function ,the important plasma electrolytes are (Na^+ , K^+ , Cl^- , H^+ , Ca^{++} , Mn^{++}).



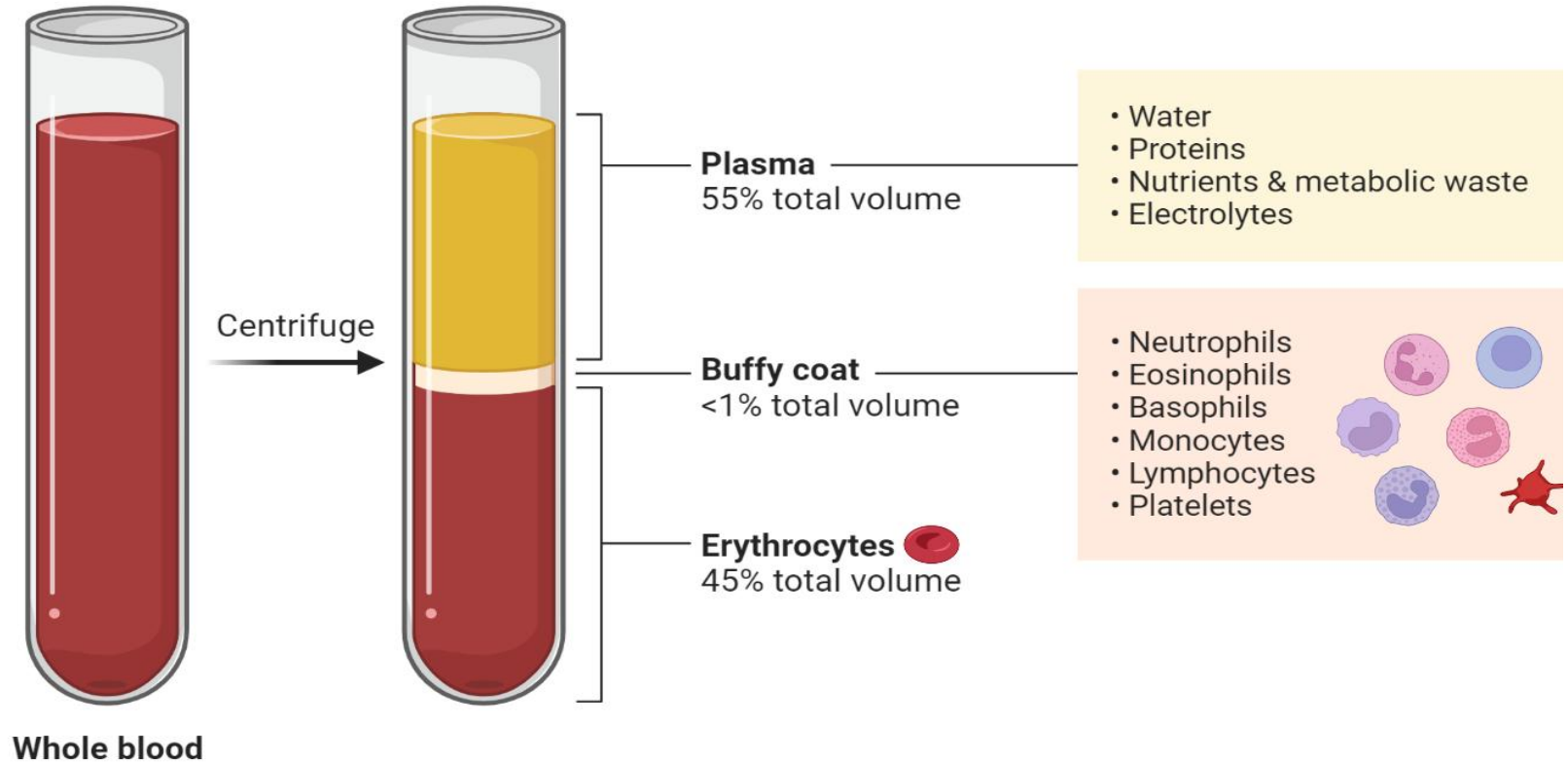
Blood

3. Proteins:

- Coagulation protein (fibrinogen)
- Immunoproteins (antibody)
- Transport proteins (Albumin, Transferrin).

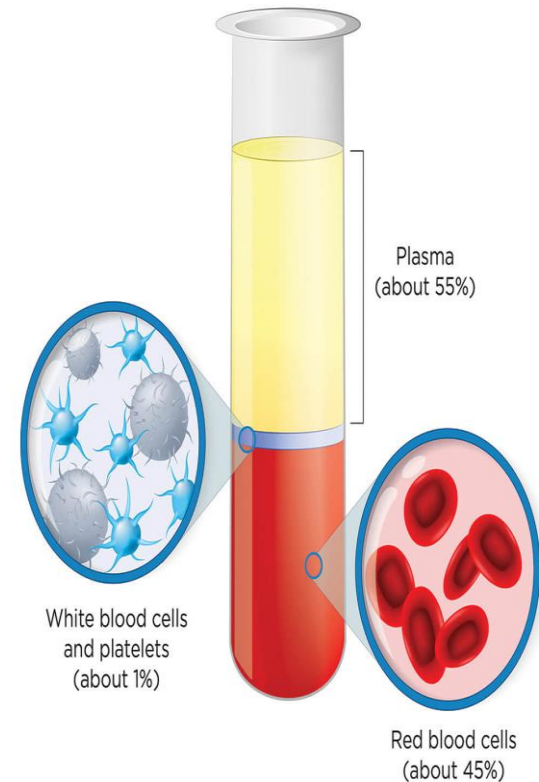


Blood and Its Components



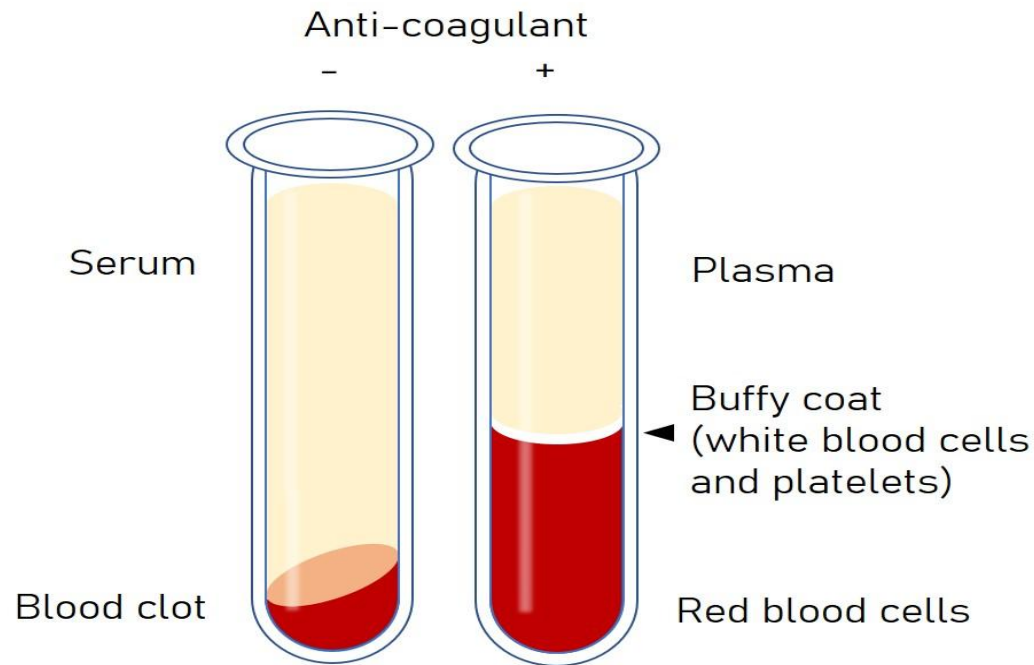
Blood plasma and serum

- **Plasma:** is the fluid portion of the blood in which cellular elements are suspended and circulated through the body.
- **The volume of plasma part** of blood is usually a little more than half the volume of whole blood volume.
- **Normal volumes:** 2.6 L, blood cell 2.4 L, total blood volume 5L).



Blood plasma and serum

- **Serum:** is the clear fluid that separates from blood upon coagulation when all cellular elements are removed.
- **Serum** = Plasma – Fibrinogen



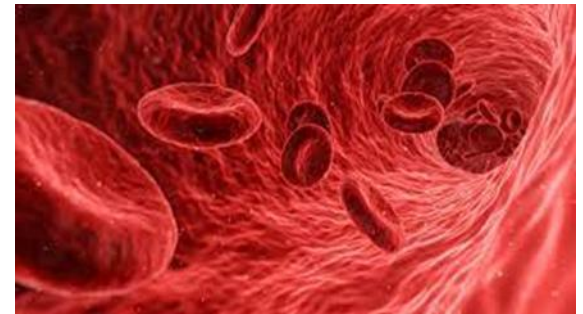
Anticoagulants

1- EDTA Na^+ or K^+ salts (Ethylene diamine Tetra Acetic Acid) used for hematological work.

2- Tri- Sodium Citrate used for coagulation studies.

3- Heparin used for fragility test.

4- Acid Citrate Dextrose used for blood transfusion.



Blood Collection



Blood Collection

- Blood can be collected from 3 different sources:

1- Capillary blood



2- Venous blood

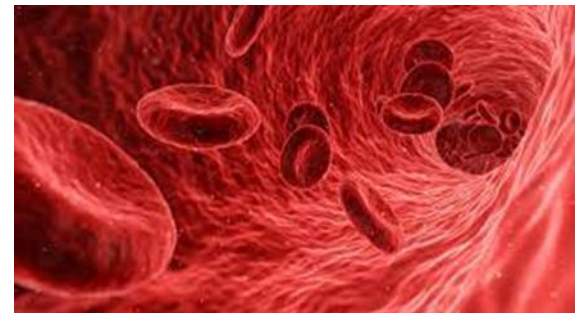


3- Arterial blood



Blood Collection

- **Blood samples** will be collected for experiments from the subjects themselves.
- Most of the experiments would require only a drop of blood which could be obtained conveniently through **a puncture at the tip of a finger.**
- When more blood is needed, it is collected by **puncturing a vein** by a needle and syringing the required blood.



Blood Collection Tools

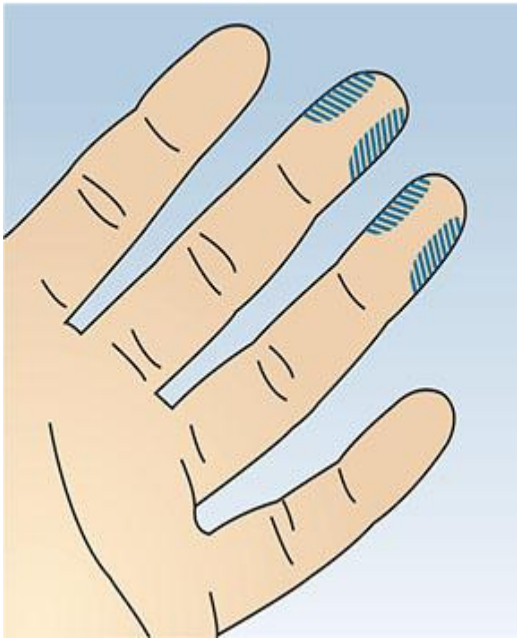
- **Material and Tools:**

- Tourniquet.
- Vacutainer or syringe.
- Alcohol swab.
- Bandage/ medi-plast.

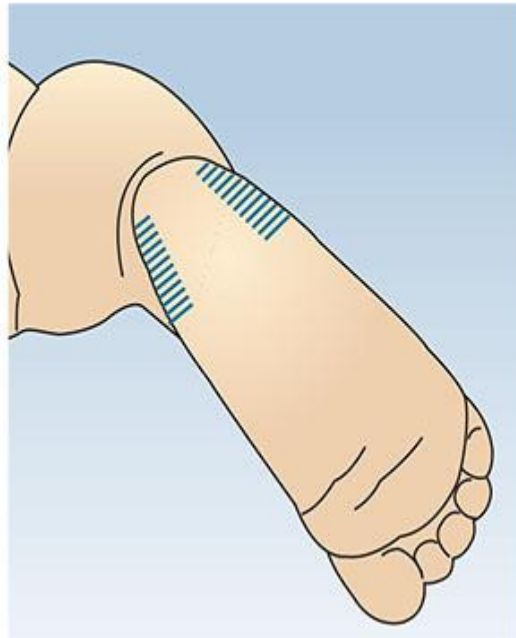


Capillary blood Collection site

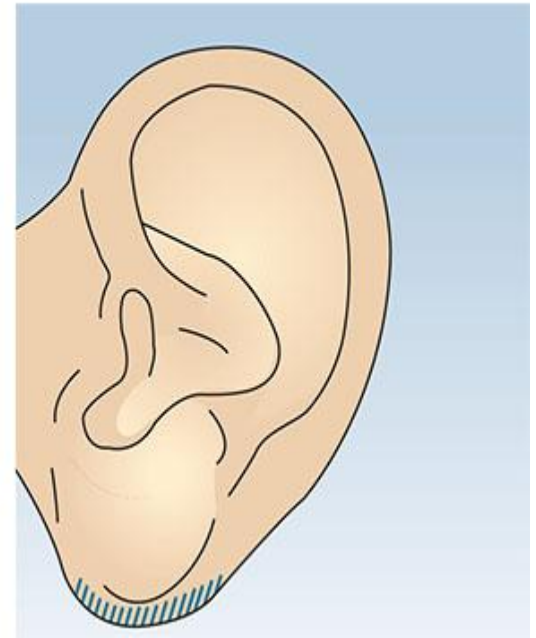
① Fingertip



② Heel

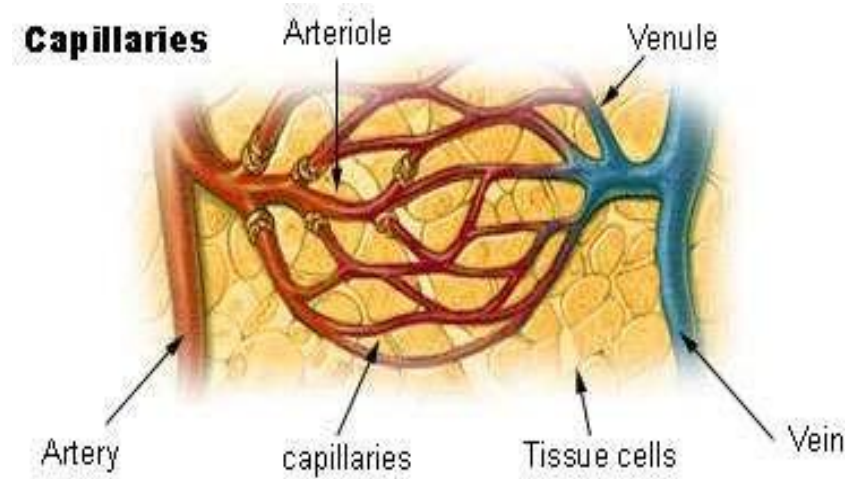


③ Earlobe



1- Finger (Capillary puncture):

- When a finger is punctured, blood in the capillaries will ooze out. This blood should be collected and utilized quickly, before clotting as there are no anticoagulants added, sometimes special heparinized capillary tubes may be used for special investigations.
- **From infants** blood sample may be collected conveniently from the heel or the big toe



Finger (Capillary puncture) Procedure

- 1- Sterilize the skin thoroughly over and around the site to be punctured with a cotton pad soaked with suitable disinfectant (70% alcohol).
- 2- Wipe the area with dry, sterile gauze pad.



Perform hand hygiene



Put on well-fitting gloves



Wipe the fingertip with the alcohol pad

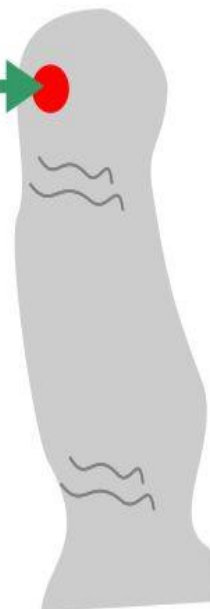
Finger (Capillary puncture) Procedure

3- Puncture the skin with a sterile lancet. Do a quick, single and sufficiently deep stroke to puncture to allow free flow of blood. Flow of blood could be enhanced by gentle pressure at the proximal part of the finger to obstruct venous return. Squeezing or milking the finger will dilute the blood with tissue fluid .



Finger (Capillary puncture) Procedure

- Finger prick
- 2 to 3 mm deep
Midway between the edge and midpoint of the finger
- Should not be near the nails
- Avoid excessive squeezing

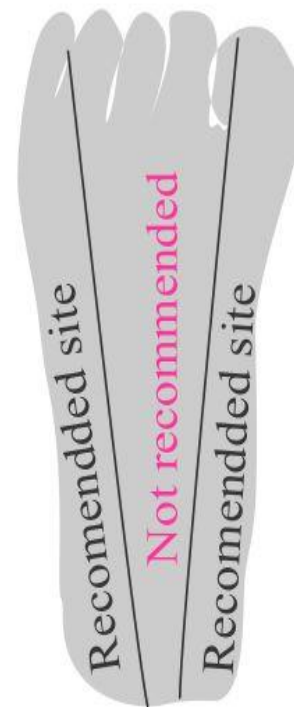


Correct



Incorrect

- Best for the diabetics
- Avoid ear lobe because the blood flow is slower
So may get raised Hb



- Don't take blood from the back of heel

Finger (Capillary puncture) Procedure

4- Allow the blood to flow freely and discard the first drop by wiping it away with a dry gauze pad .

5- Obtain the required amount of blood directly using appropriate instrument, pipette or slide.



Puncture fingertips with an appropriate blood collection device



Wipe away first drop of blood



Collect blood into the capillary



Position the tube vertically

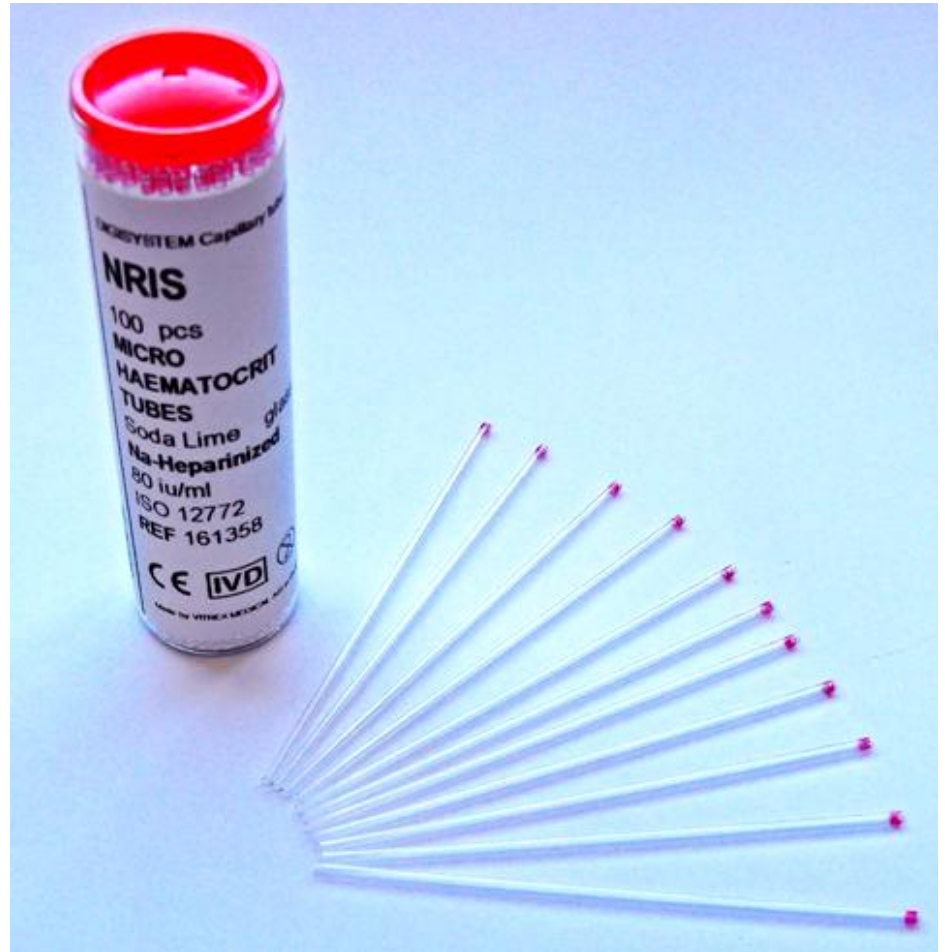


Gently inverse 3-5 times



Get Blood sample

Capillary tube



Finger (Capillary puncture) Procedure

6- Leave the punctured site undisturbed for the bleeding to stop on its own and if it continues to bleed, cover the area with clean gauze and apply gentle pressure until bleeding stops.



2- Venipuncture procedure

- **Venous blood** can be conveniently obtained from the veins in the bend of the elbow. A tourniquet applied above the site of collection facilitates blood collection by obstructing the venous return while permitting arterial flow into the site.



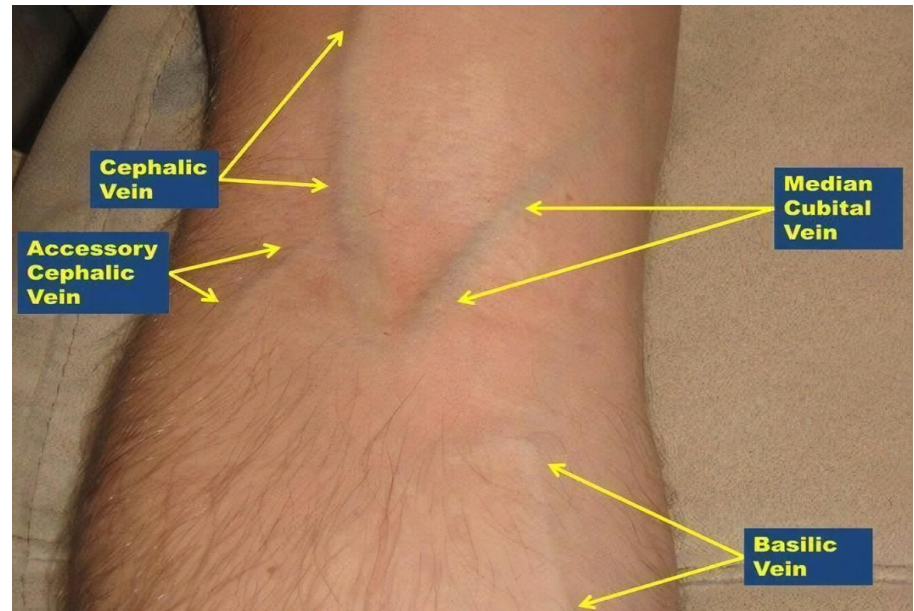
2- Venipuncture procedure

- Other sites may be selected to collect blood from children and dehydrated or collapsed patients. Syringe of appropriate size could be used. The needle should be sterile, sharp and not smaller than 20 gauge in diameter



2- Venipuncture procedure

1- Inspect the forearm and identify the vein to be punctured.



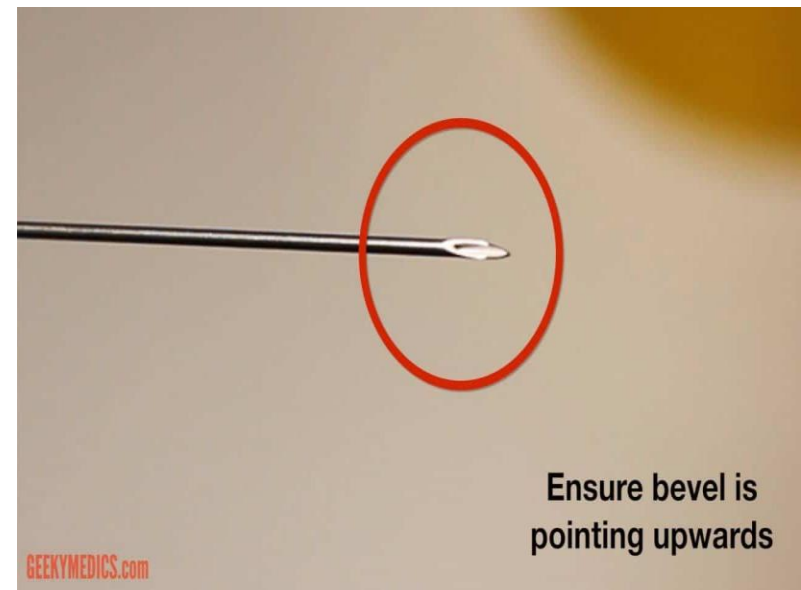
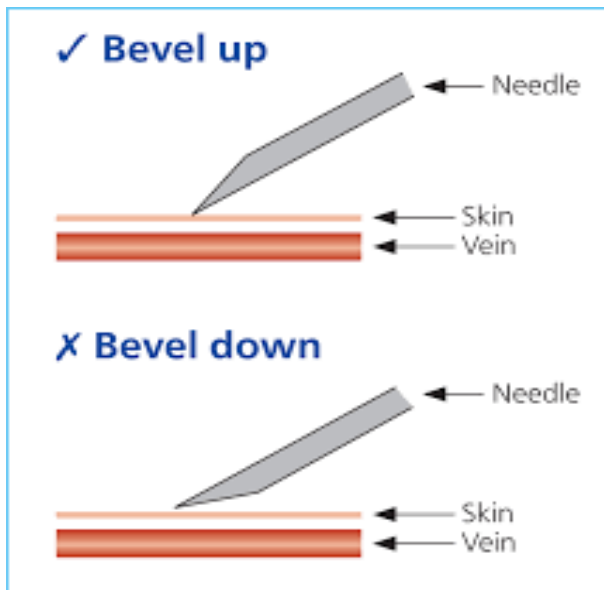
2- Venipuncture procedure

2- Sterilize the skin over and around the site to be punctured with a cotton pad soaked with suitable disinfectant (surgical spirit- 70% alcohol).



2- Venipuncture procedure

- 3- Wipe the area with dry, sterile gauze pad .
4. Prepare the syringe and needle carefully following aseptic techniques. Make sure that there is no air in the syringe.



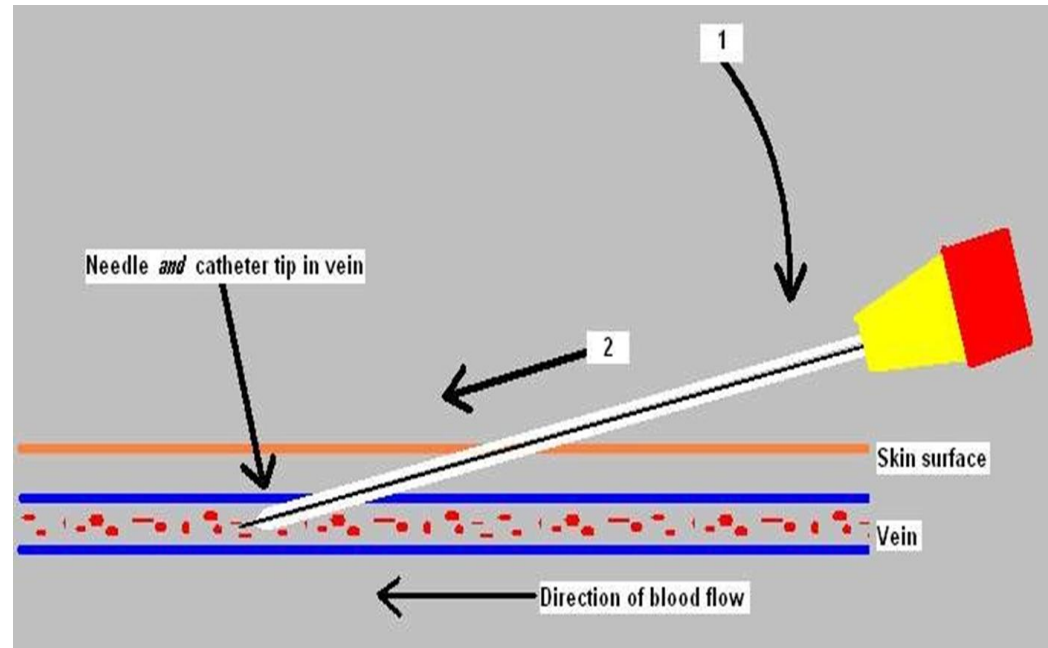
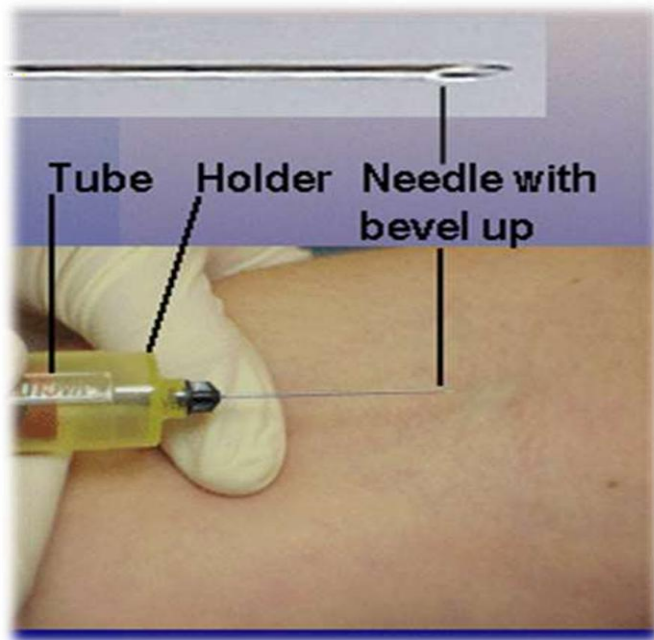
2- Venipuncture procedure

5. Puncture the skin, with the needle attached to the syringe, just closer to the vein [not above the vein] and push the needle for about 2 – 3 mm under the Skin.



2- Venipuncture procedure

5. Puncture the skin, with the needle attached to the syringe, just closer to the vein [not above the vein] and push the needle for about 2 – 3 mm under the Skin.



2- Venipuncture procedure

6. Fix the syringe and draw the necessary amount of blood .
7. Release the tourniquet first and then place dry gauze over the puncture site, exert a gentle pressure and draw the needle out. Ask the subject to hold the gauze for about five minutes to avoid oozing of blood from the puncture site.



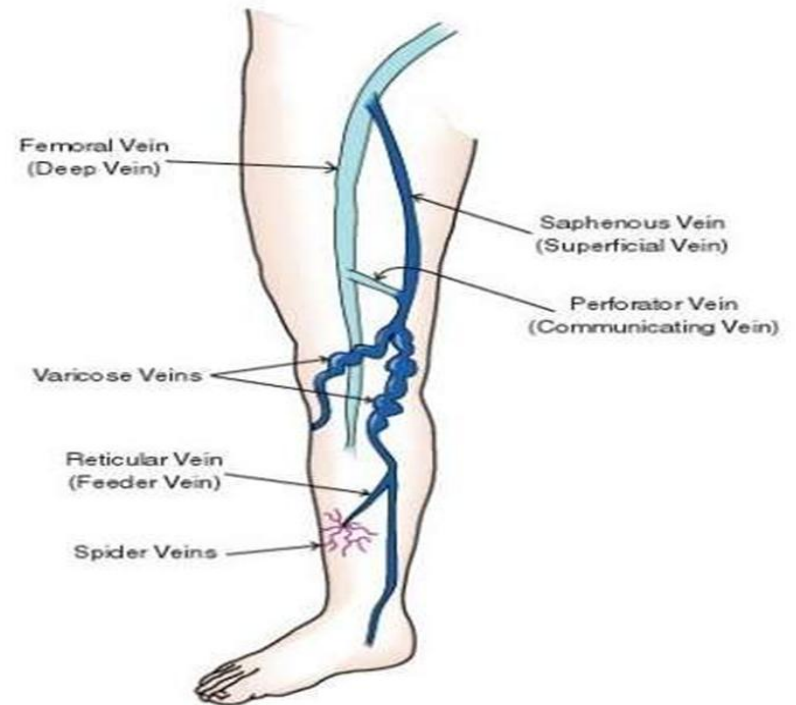
2- Venipuncture procedure

- If difficult to obtain from the ante-cubital fossa we can draw blood from following various site:

1- Dorsum Vein

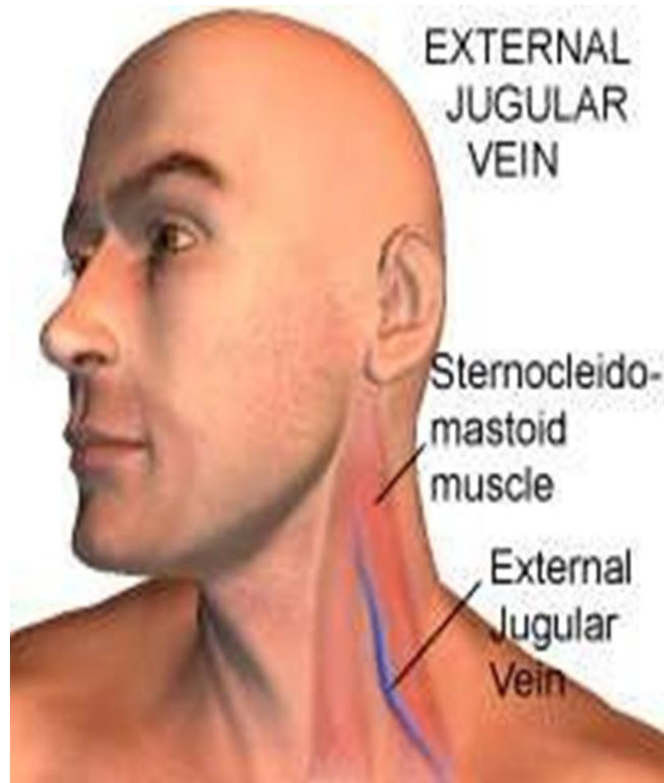


2- Femoral Vein

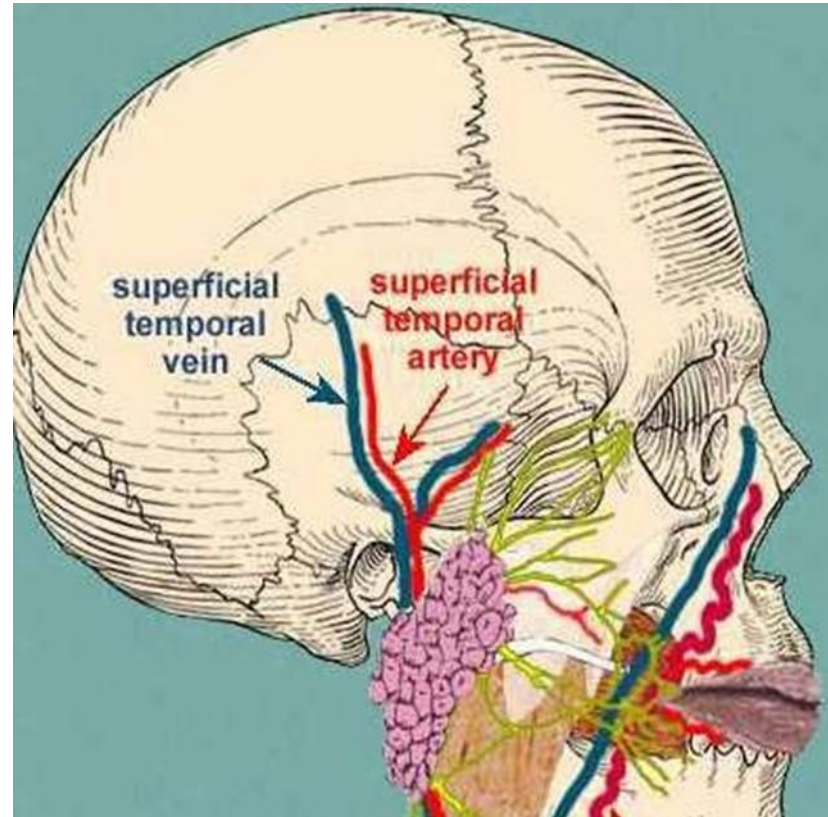


2- Venipuncture procedure

3- Jugular vein



2- Scalp vein



2- Venipuncture procedure

Precaution:

Venipuncture area must be cleaned/sterilized properly. Tourniquet should not be applied for a long time and not more than 1 min.

To avoid stasis of blood



Stasis



concentration

RBC

haemoglobin

albumin

calcium

3- Arterial blood

- Specially required for estimation of blood gases (ABG): PH, CO₂ and O₂

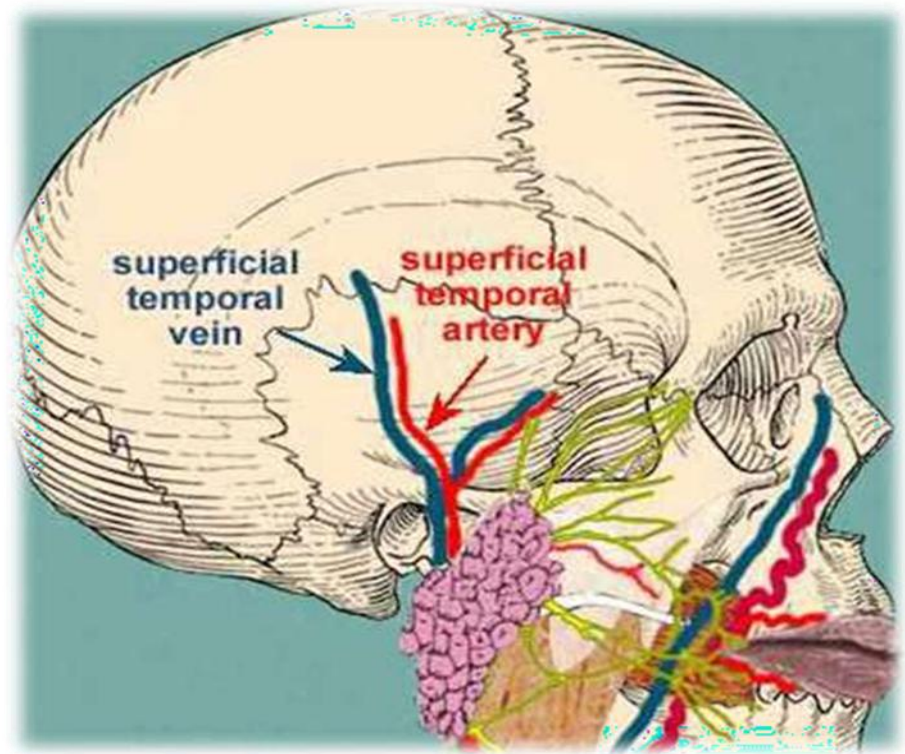
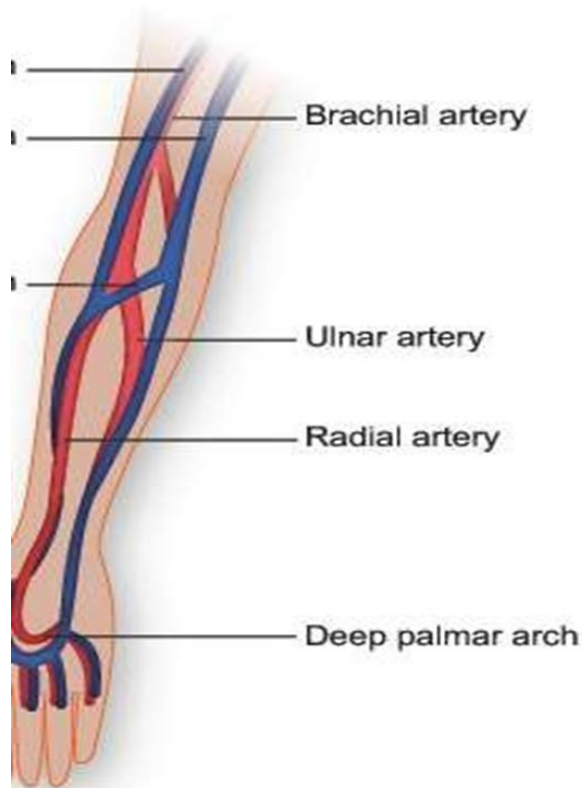


3- Arterial blood

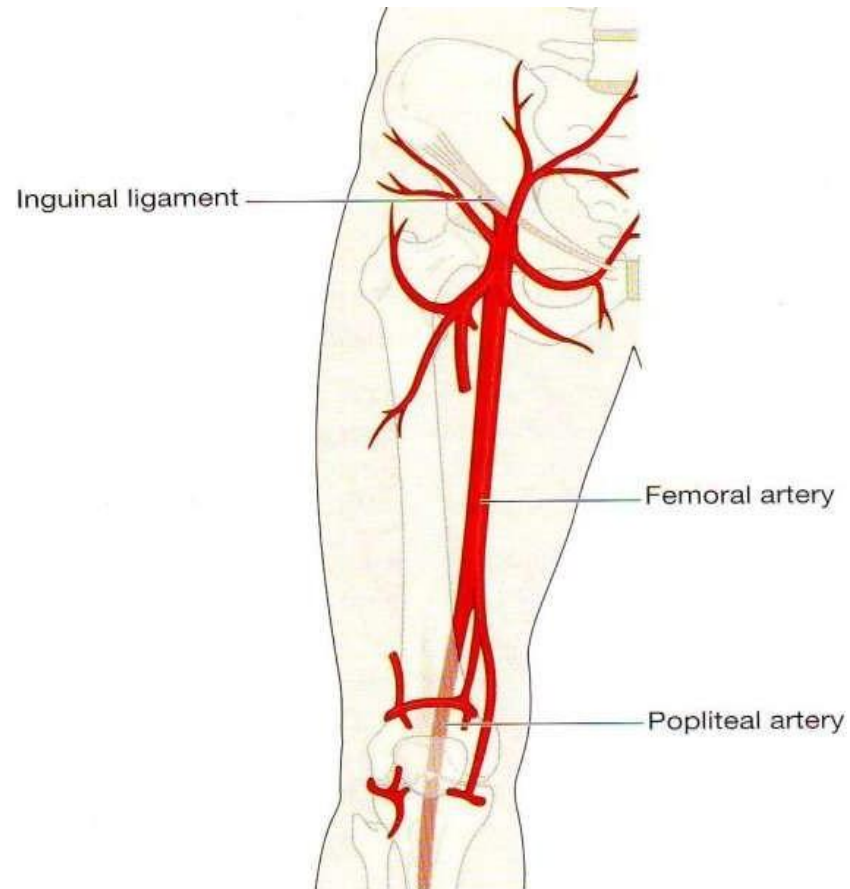
- Arterial blood can be obtained from a superficial artery such as:

1. Radial and brachial

2. Temporal artery



3. Femoral artery



The femoral artery and its main branches.

Blood Collection Tubes

- Plastic tube with a rubber stopper include color coded.
- Contain anticoagulants and/or other chemical additives.
- Plain tubes contain no anticoagulants.
- All tubes must be mixed thoroughly.



Anticoagulant tube

- **EDTA** (Ethylene Diamine Tetra-Acetate) liquid:
 - **Types:** Na and K₂ EDTA (1.5-0.25mg /ml)
 - **Functions** by forming Ca salts to remove Ca.
 - **Uses:** most hematology studies. such as:
CBC, PCR and HbA1c.
 - Requires full draw (invert 8 times).



Light Blue :

- Sodium citrate (1:9 ratio).
- Anticoagulant: 32g/l.
- Action: Remove Calcium.
- Uses: Coagulation studies and platelet function.



Dark Green

- **Sodium Heparin** or Lithium Heparin anticoagulant.
- **Action:** inactivate thrombin and thromboplastin.
- **Uses:**
 - For Lithium level use Na Heparin anticoagulant
 - &for Ammonia level use Na or Lithium Heparin



Red (Plain tube):

- No preservative/anticoagulant.
- **Uses:** usually for blood bank tests, toxicology and serology



SST/ Gold top tube

- SST (Serum Separator Tube)
- No additives.
- Clotting accelerator and separation gel.
- **Uses:** Chemistry, Immunology, and Serology.



PST /Light Green

- Plasma Separating Tube with Lithium Heparin
- **Uses:** Chemestries



Black tube:

- **Na citrate** 1:4.
- **Action:** Remove calcium.
- **Uses:** Westergren sedimentation rate (ESR).

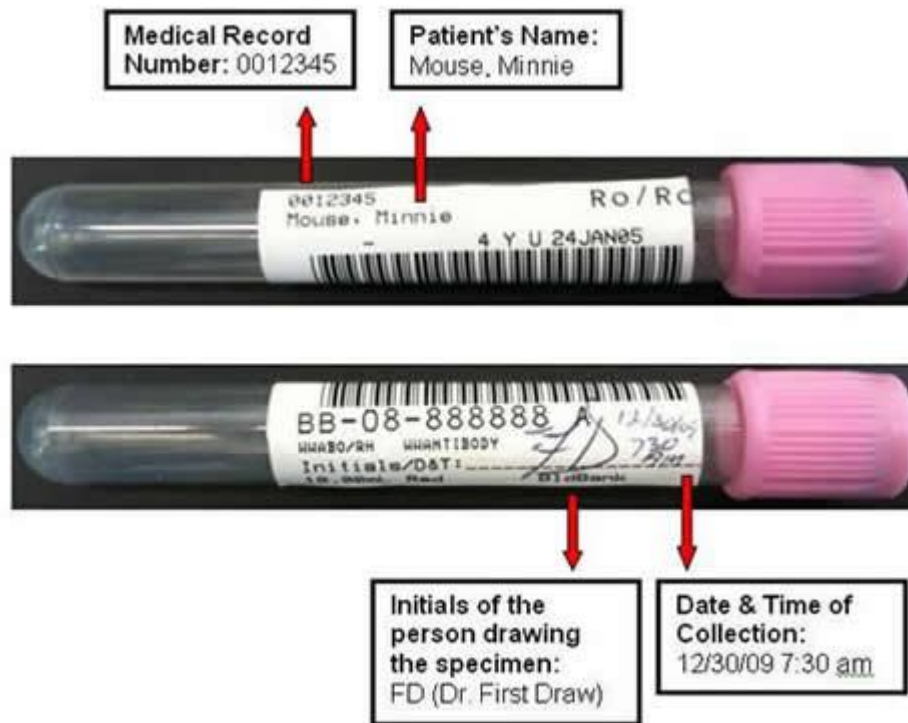


- **ESR tube**
- **Additive:** 3.8% sodium citrate



Blood Collection Tube

Note: all blood samples should be clearly labeled with **patient's name**, **number**, **time of collection** and **the date**.



Blood Collection Tube

DO



DON'TS

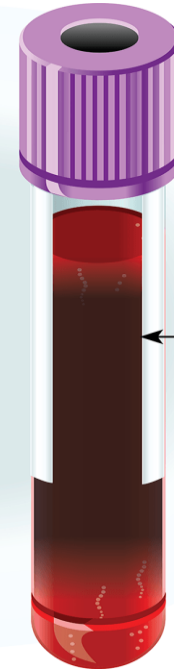


Place label
directly
under cap

Name at
the top

Barcode
straight

Collector's
User ID



Leave visible
window to
see blood



Thank you!